

Test-time Generalization for Physics through Neural Operator Splitting

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TL;DR Neural operators fail out-of-distribution. We generalize **at test time, with frozen weights**, by searching over compositions of pretrained operators and combining them via **neural operator splitting**.

The problem

Neural surrogates fail to generalize out of distribution — and existing paradigms fall short: pretrain then fine-tune needs target samples; meta-learning is costly and unstable at adaptation; both lack **compositionality**.

We study **zero-shot generalization** of neural operators, given only a few snapshots of the system. **Training** follows a parametric setting

$$\partial_t u = \sum_{k=1..K} \mu_k \mathcal{F}_k(u, \nabla u, \nabla^2 u, \dots)$$

where each trajectory has a **single class of active operator** — e.g. advection or diffusion, never both. **At test time**, we consider two zero-shot OOD settings:

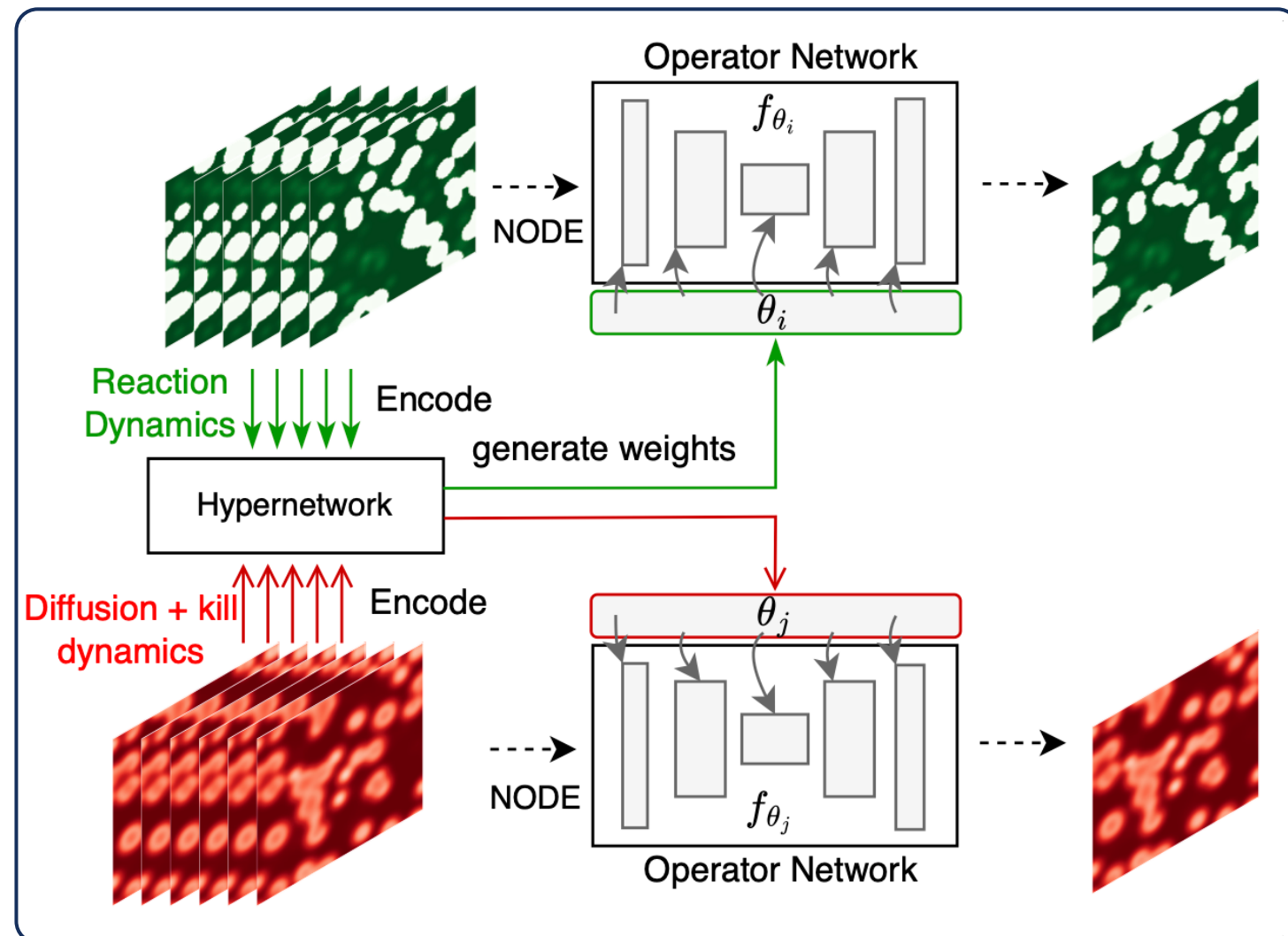
Extrapolation — a coefficient μ_k pushed outside its training range,

Composition — several operators active at once: physics never seen in training, and propose a method that improves OOD generalization in this context.

Method — neural operator splitting at test time

1 Pretraining

A hypernetwork (DISCO) encodes each training trajectory into a neural-ODE operator — a **dictionary** $\{f_1, \dots, f_N\}$.



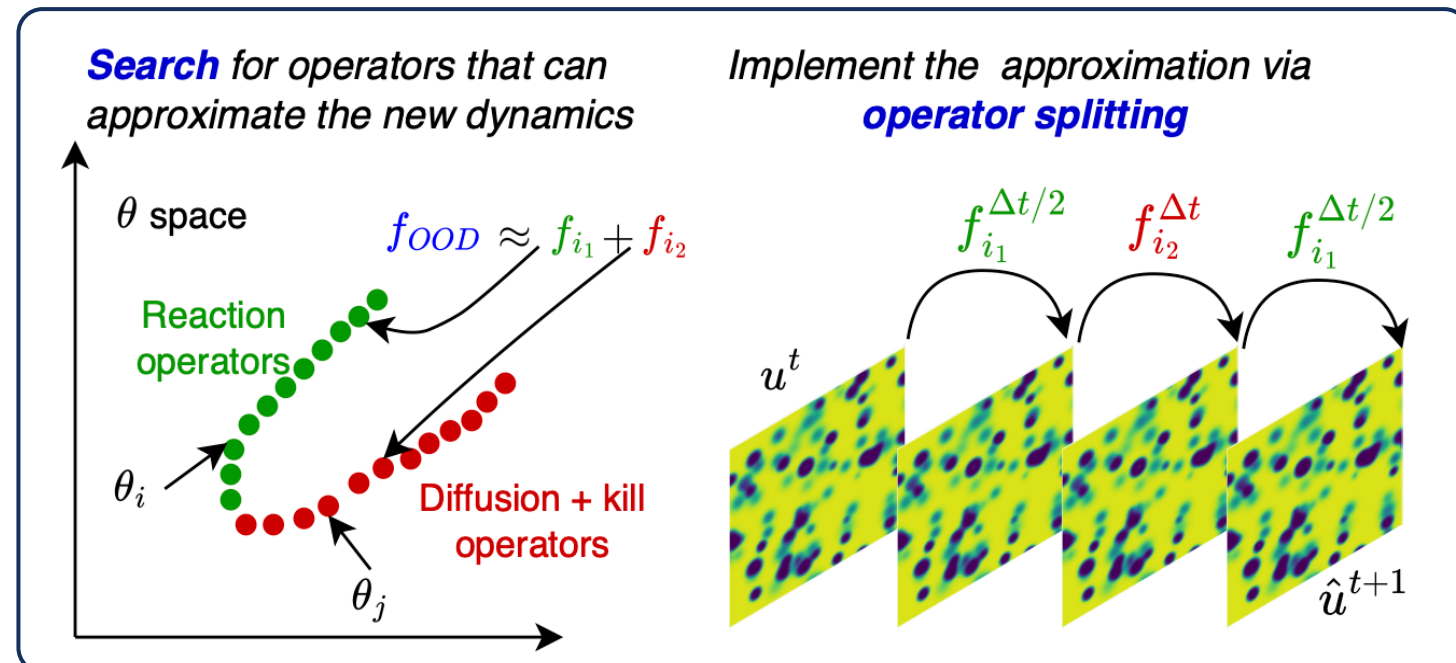
2 OOD setting

A test trajectory with **unseen dynamics** f_{OOD} and a short observation window.



3 Test-time strategy

Search for the operators that best fit the window (uniform or beam), then roll out via Lie / Strang **splitting**.



Physics composition (NRMSE ↓)

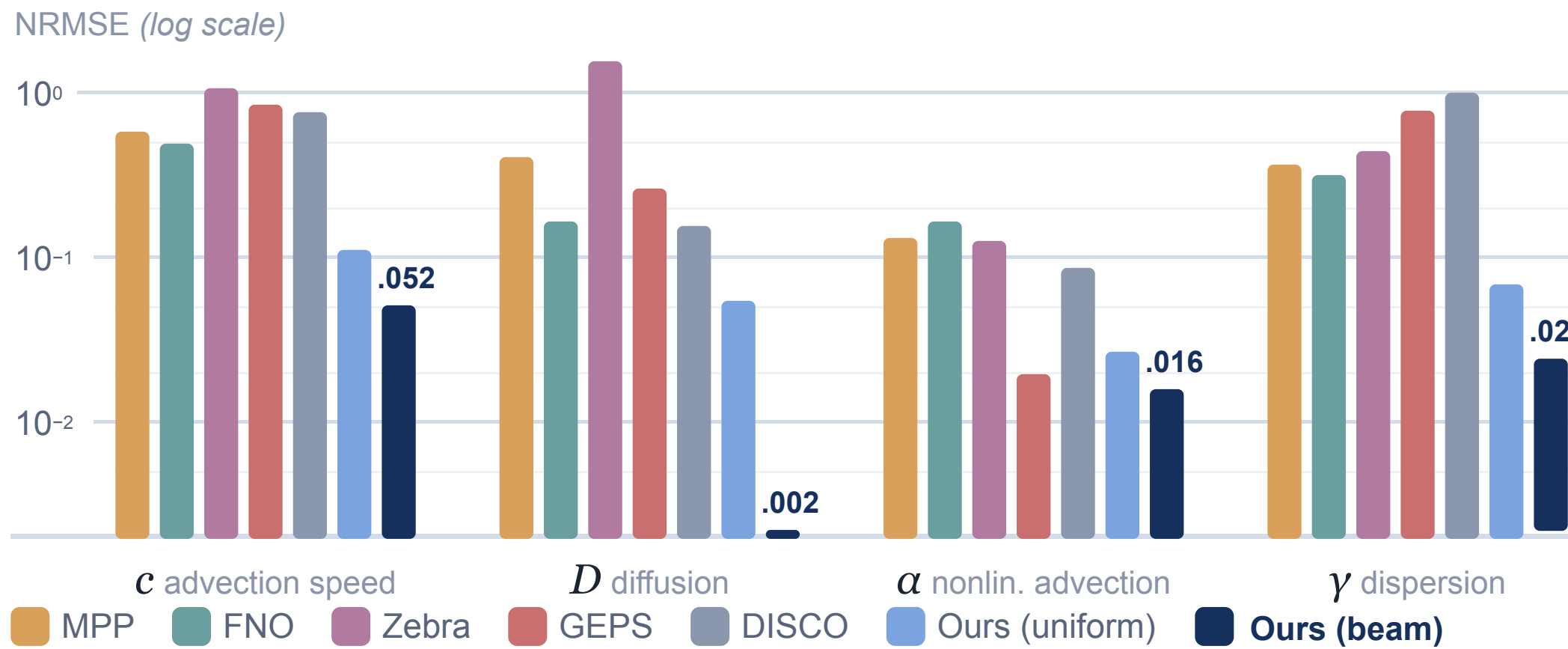
Unseen combinations of operators seen only in isolation during training.

Method	Adv + Diff	NL adv + Diff	NL adv + Disp	Diff + Disp	All three	Reac + Diff	Euler + Diff
MPP	0.270	0.050	0.105	0.091	0.128	0.191	0.273
FNO	0.318	0.031	0.165	0.038	0.129	0.224	0.241
Zebra	0.893	0.022	0.241	0.069	0.193	0.127	0.198
GEPS	0.039	0.039	0.249	0.229	0.265	0.128	0.786
DISCO	0.170	0.085	0.100	0.120	0.164	0.245	0.572
Ours (unif.)	0.043	0.068	0.103	0.043	0.075	0.089	0.209
Ours (beam)	0.015	0.056	0.049	0.007	0.036	0.089	0.066

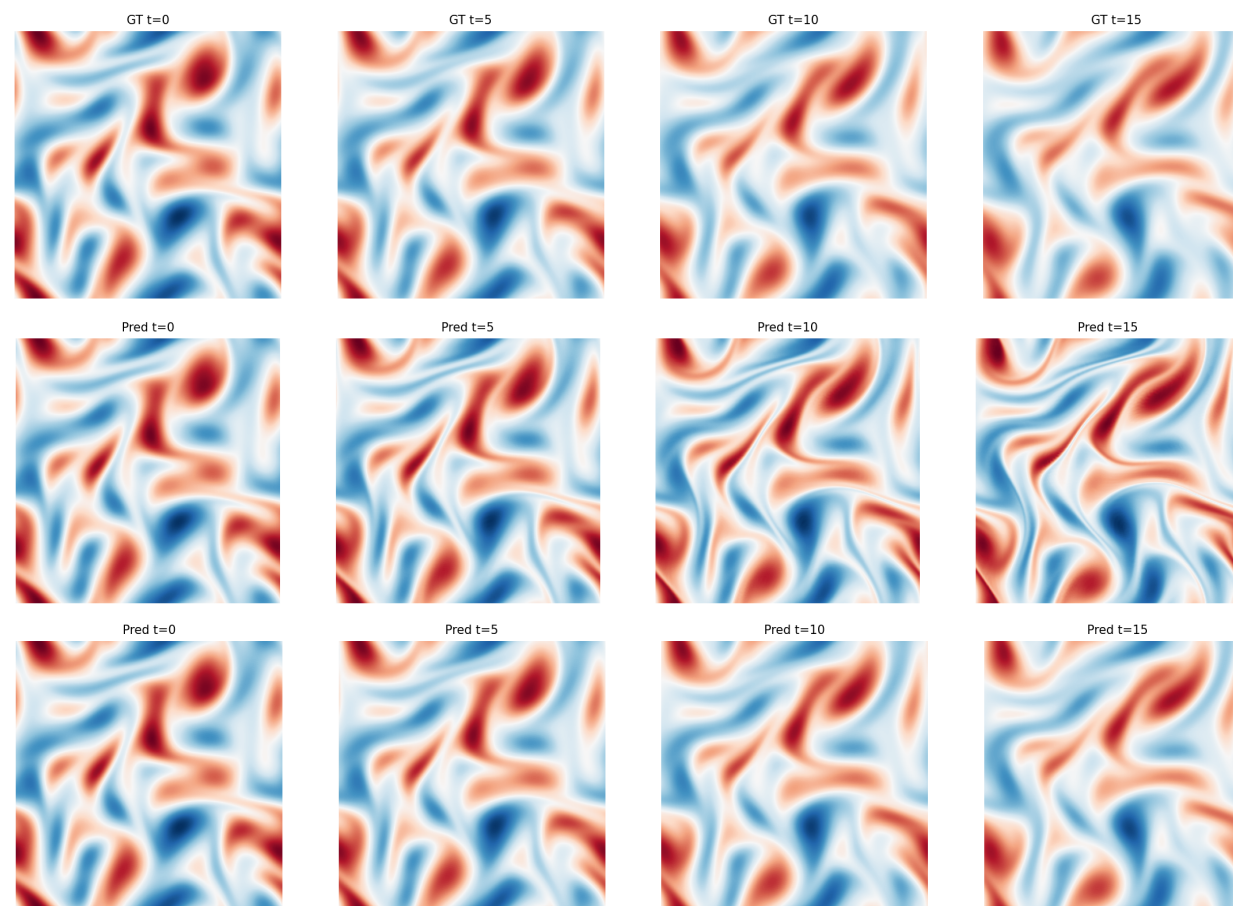
Bold = best. Beam search wins on the hardest coupled physics, including Navier–Stokes (Euler + Diff). Adv — advection · NL adv — nonlinear advection · Diff — diffusion · Disp — dispersion · Reac — reaction

Parameter extrapolation (NRMSE ↓)

Coefficients pushed *outside* the training range.

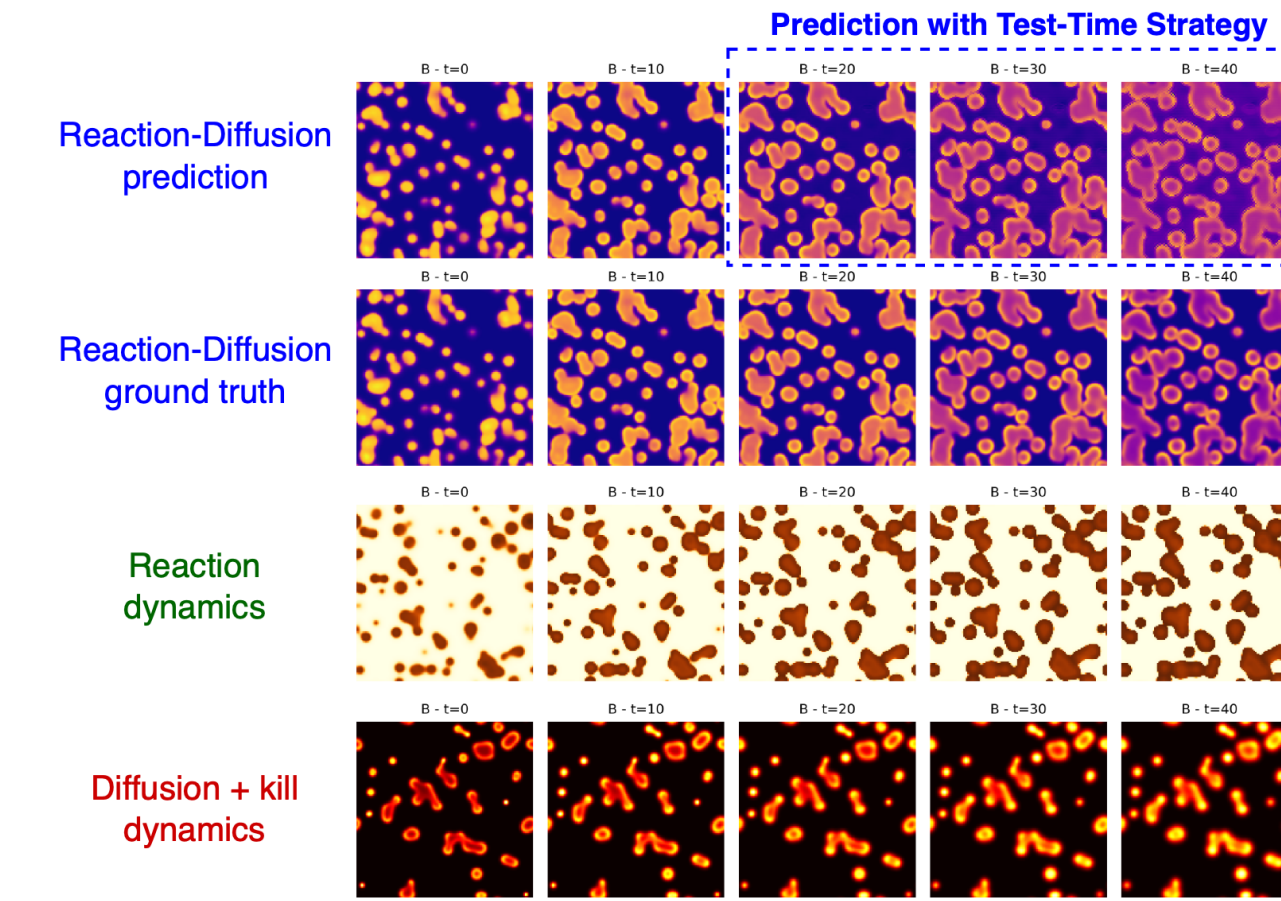


Zero-shot Navier–Stokes ($\nu = 10^{-2}$)



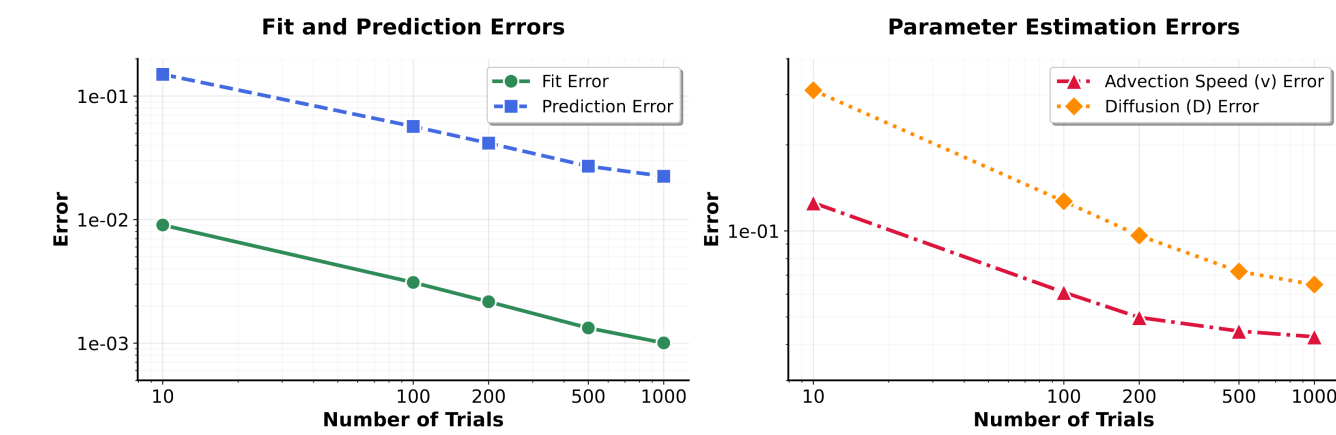
Composing **Euler + diffusion** — never trained on Navier–Stokes — cuts the rollout error **4× here, ~9× on average** vs. direct DISCO encoding.

Zero-shot Gray–Scott reaction–diffusion



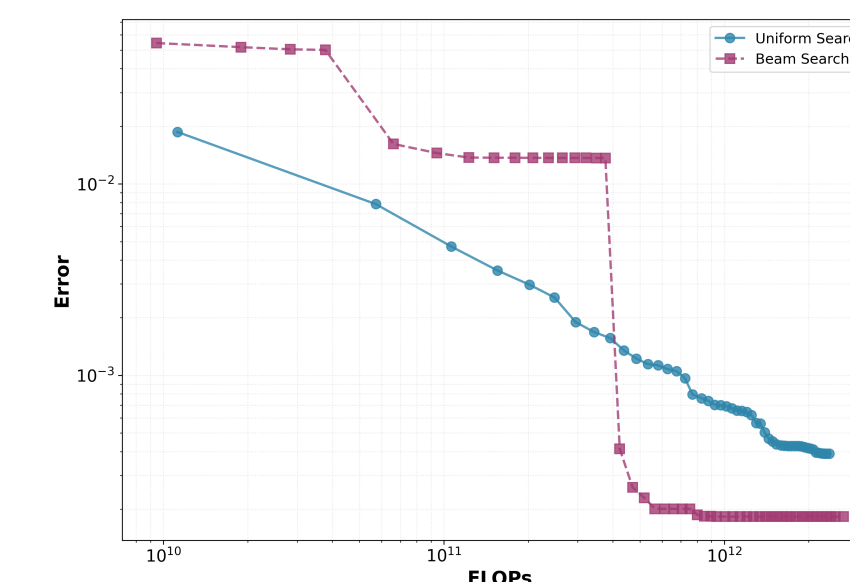
Reaction and diffusion operators, seen *only in isolation* during training, recombined at test time to predict coupled dynamics.

Test-time compute improves accuracy



More search trials → lower fit & prediction error (left), accurate recovered PDE parameters (right) — **system identification for free.**

Beam search is more compute-efficient



Fitting error vs. cumulative compute (FLOPs) on **OOD advection extrapolation**. Beam search grows compositions progressively — singles, pairs, triples — dropping an order of magnitude below uniform sampling.

vs. gradient-based adaptation (per-sample NRMSE / wall-clock, A100)

Method	Adv+Diff	Adv. extrap.	Diff. extrap.
GEPS · 2000 steps	0.034 / 57 s	diverges	diverges
Ours (beam)	0.015 / 21 s	0.052 / 26 s	0.002 / 26 s

More gradient steps **hurt** GEPS far-OOD; more search compute **monotonically helps** ours.

Limitations & future work

Composed operators must share compatible input/output domains, and search cost may bottleneck in 3D. Next: broader compositionality across **spatial dimensions, discretizations, and physical fields** — and scaling the search efficiently to 3D.

References

DISCO — Morel, Han, Oyallon. *Learning to Discover an Evolution Operator for Multi-physics-agnostic Prediction*. ICML 2025. **FNO** — Li et al. ICLR 2021. **MPP** — McCabe et al. 2023. **Zebra** — Serrano et al. ICML 2025. **GEPS** — Kassāf Koupaī et al. NeurIPS 2024. **Operator splitting** — Strang. SIAM 1968.